STATS 369 Project: Predicting Public Sentiment and Behaviour on Digital Wellbeing in Asia

Total: 40 points

Lisa Chen

2026-04-24

- 1 Project Overview
- 2 The Brief from Your Client
- 3 Requirements
- 4 What do you need to do?
  - 4.1 Gather Project Background and Scope
  - 4.2 Create an Example Dataset
  - 4.3 Database Integration
  - 4.4 Feature Engineering and Preprocessing
  - 4.5 Modelling and Prediction
  - 4.6 Report Writing [40 marks]
- 5 The Intended Audience for the Report
- 6 Assessment Criteria
- 7 AI Usage and Academic Integrity

1 Project Overview

In this project, you will develop a **complete data science workflow**, from data creation and database integration to modelling and prediction, and communicate your findings through a structured report. The work must explicitly consider ethical, cultural, and societal implications.

2 The Brief from Your Client

You have been engaged by a regional digital policy unit in Asia, referred to as DPU. The organisation is exploring how data science and AI models can support policy decisions around:

- digital wellbeing, such as excessive screen time;
- online civility and harassment;
- misinformation and trust; and
- cultural norms in online interactions, such as indirect communication and respect hierarchies.

DPU is considering building a system that:

1. classifies sentiment as positive, neutral, or negative;
2. predicts risk levels, such as the likelihood of harmful or toxic discourse; and
3. provides aggregated insights to inform policy or platform interventions.

The system would process hundreds of thousands of user comments annually, sourced from forums, surveys, and social media platforms, accessible via CSV files or SQL databases.

Your role is to conduct a scoping and prototype study to demonstrate feasibility, modelling approaches, and associated risks.

3 Requirements

Your submission must include:

- a sample dataset;
- an R Markdown (`.Rmd`) file; and
- a well-structured final report, in PDF or HTML format, presenting the reproducible analysis conducted on the sample dataset.

4 What do you need to do?

4.1 Gather Project Background and Scope

The success of a data science project begins with a clear understanding of the background, context, and potential impact of the problem.

You should consider the relevant context for understanding public sentiment on social media use and wellbeing—reflect on perspectives from yourself, your peers, and other diverse groups. Define the scope of the project, including what will be included and what will be excluded. Clearly state the project's goal and objectives, as these will set the direction for your report.

4.2 Create an Example Dataset

Construct a tidy dataset with the following columns:

- `platform` , for example forum, survey, or social media;
- `year` , e.g. from 2020 to 2025.
- `topic` , for example wellbeing, misinformation, or online behaviour;
- `comment_text` , for example "I think people should be more respectful online.", "It can influence behaviour over time.";
- `label_sentiment` , with ground-truth labels: positive, neutral, or negative; and
- `label_risk` , for example low, medium, or high risk.

The dataset should meet the following requirements:

- at least four platforms;
- at least three topics;
- at least 500 observations; and
- each comment should contain two to three sentences and should be culturally contextualised, with Asian social and cultural norms encouraged.

You may use AI tools to generate realistic sample data, but you should review and edit the comments to ensure they are appropriate, coherent, and ethically suitable for use in the project.

4.3 Database Integration

Demonstrate how to:

- create and connect to a suitable database (such as DuckDB);
- load the dataset into the database; and
- query the data for analysis.

Briefly justify:

- why databases are useful at this scale; and
- the trade-offs between using databases and flat files such as CSV files.

4.4 Feature Engineering and Preprocessing

Prepare the data for modelling. You should consider:

- text preprocessing, such as tokenisation and stopword removal;
- feature extraction, such as TF-IDF or n-grams; and
- cultural and linguistic considerations, such as indirect tone, mixed-language comments, or multilingual text.

Clearly explain and justify your choices.

4.5 Modelling and Prediction

Develop and compare at least two predictive models. Suitable examples include:

- logistic regression or regularised regression models;
- tree-based models, such as random forests or gradient boosting; and
- simple NLP-based classifiers.

Your modelling work should:

- predict sentiment and/or risk level;
- evaluate performance using appropriate metrics, such as accuracy, precision, recall, or a confusion matrix;
- compare models and justify your preferred approach;
- discuss overfitting and generalisation;
- provide a reasonable estimate of prediction error; and
- demonstrate predictions on new or unseen data.

4.6 Report Writing [40 marks]

Prepare a clear and structured report of **no more than 15 A4 pages**, excluding the cover page. You report should include the following section headings.

4.6.1 Title and Author [2 marks]

Include an appropriate title and your name as the author – this can be your cover page of the report.

4.6.2 Executive Summary [4 marks]

Write an executive summary of **no more than 200 words**. It should summarise:

- the purpose of the study;
- the analytical approach;
- key findings; and
- high-level recommendations.

4.6.3 Introduction [4 marks]

The introduction should:

- explain the problem context, focusing on digital wellbeing and online behaviour in Asian societies;
- outline the key questions investigated;
- briefly describe the dataset; and
- include a database subsection explaining why database systems are relevant.

4.6.4 Data and Approaches [10 marks]

The approaches section should describe:

- data generation and structure;
- database integration;
- feature engineering and preprocessing;
- modelling approaches; and
- the evaluation framework.

4.6.5 Results [5 marks]

The results section should include:

- model performance comparisons;
- key predictive insights; and
- example predictions.

4.6.6 Summary and Discussion [10 marks]

The discussion section should include the following subsections.

4.6.6.1 Summary

Summarise the main findings and their implications.

4.6.6.2 Strengths and Limitations

Discuss both technical and data-related limitations, including:

- model limitations, such as bias, interpretability, and generalisation;
- data limitations, such as the use of synthetic data and possible lack of representativeness; and
- limitations of sentiment and risk classification in culturally nuanced settings.

4.6.6.3 Ethical Considerations

You must explicitly address:

- bias and fairness, including cultural bias in language and sentiment interpretation;
- privacy and consent, especially when using public versus private comments;
- risks of misuse, including surveillance, censorship, or social scoring;

- cultural sensitivity, including indirect communication styles common in many Asian contexts; and
- transparency and accountability in the use of models for policy decisions.

4.6.6.4 Recommended Next Steps

Briefly outline suggested next steps, such as:

- improving the model using multilingual or contextual NLP approaches;
- strengthening data governance;
- conducting a small-scale pilot study; and
- reviewing the system with relevant stakeholders before any real-world deployment.

4.6.7 References [5 marks]

Include appropriate references, such as:

- R packages used in the analysis;
- database and modelling resources;
- relevant methodological references; and
- ethical frameworks or guidance on responsible data science and AI.

You should include in-text citations throughout the report where appropriate. Package citations can be found using the `citation()` function in R.

5 The Intended Audience for the Report

Your report should be written for two audiences:

- policy stakeholders, who may not have a technical background; and
- technical reviewers, such as data science or statistics experts.

The Executive Summary, Introduction, and Discussion should be accessible to non-technical readers. The Methods and Results sections should provide sufficient detail for technical review.

6 Assessment Criteria

This project assesses your ability to:

- apply an end-to-end data science workflow;
- build and evaluate predictive models;
- communicate clearly through a structured report;
- demonstrate ethical judgement; and
- show cultural awareness in an Asian context.

7 AI Usage and Academic Integrity

The use of AI tools is permitted in this project; however, their use must be **responsible, transparent, and appropriately acknowledged**.

You are responsible for:

- critically reviewing and verifying any AI-generated content;
- ensuring that all outputs are accurate, relevant, and appropriate for the task; and

- integrating AI-assisted material into your work in a thoughtful and academically sound manner.

All use of AI must be clearly declared/referenced. This includes, where appropriate:

- identifying the AI tool used (e.g. ChatGPT or other platforms), describing how it was used (e.g. idea generation, drafting, data simulation), and
- ensuring that AI-generated content is properly adapted and not presented as entirely original work.

You **must also uphold the academic integrity standards** of the University of Auckland. In particular:

- do not submit AI-generated content as your own without acknowledgement;
- do not copy or reproduce the work of others without proper citation; and
- ensure that the final report is written to demonstrate your own understanding and judgement.

AI should be used as a supporting tool, not a substitute for your own thinking. This project is designed to develop your capability to become an independent and responsible data scientist.